

1603

Rivers, Lakes and Drainage of India

About 1-2 questions a paper. The locked repeat is tributary-to-main-river matching (RAS 2013, 2021, 2024 - Rapti, Teesta, Hiran, Purna). Ramsar sites and the Indus Waters Treaty are the other two live zones.

Rivers are the single most reliable scoring block in Indian Geography, because the questions are always the same shape - a tributary matched to its main river, a lake matched to its state, a waterfall matched to its river. RPSC has asked the tributary match in 2013, 2021 and 2024. Build the tables once and this chapter never troubles you again.

Drainage of India

Himalayan systems

- Indus
- Ganga
- Brahmaputra

Brahmaputra

- Tsangpo
- Siang/Dihang
- Jamuna
- Majuli

Lakes

- Wular
- Vembanad
- Chilika
- Loktak
- Sambhar
- Lonar

Indus

- Bokhar Chu glacier
- Zaskar, Shyok, Nubra
- Treaty 1960
- Eastern rivers to India

West-flowing

- Narmada
- Tapi
- Mahi
- Sabarmati
- Luni

Policy

- Ramsar 1971
- NPP 1980
- NWDA 1982
- Ken-Betwa

Ganga

- Devprayag
- Panch Prayag
- Left bank - Ghaghara, Gandak, Kosi
- Right bank - Yamuna, Son

East-flowing

- Mahanadi
- Godavari
- Krishna
- Kaveri

Two families of rivers – and why they behave differently

Every Indian river belongs to one of two families, and the difference is not decorative. Himalayan rivers rise in snow, so they flow all year. They are also older than the mountains in places – they kept cutting downward while the Himalaya rose around them, which is why they slice deep gorges through the ranges. That is called an antecedent course. Peninsular rivers rise in rainfall alone, so they shrink to a trickle in the dry season, and they have been flowing so long over stable rock that their valleys are broad, shallow and nearly graded.

Himalayan vs Peninsular rivers

Himalayan rivers	Peninsular rivers
Perennial – fed by snowmelt and rain both	Seasonal – rain-fed only
Antecedent in places; cut deep gorges	Flow in broad, shallow, graded valleys
Long courses with large basins and heavy silt	Shorter courses, smaller basins, less silt
Build large deltas – the Ganga-Brahmaputra delta is the world's largest	Mostly deltas in the east; Narmada and Tapi make estuaries
Meander freely and shift course over the plains	Fixed courses on hard rock
Navigable in the plains; support canal irrigation	Mostly non-navigable; suited to dam sites

- The Ganga basin is India's largest river basin – about a quarter of the country's area.
- The Godavari basin is the largest peninsular basin; the Godavari is the longest peninsular river.
- Length order to remember: Ganga 2,525 km, Godavari 1,465 km, Krishna 1,400 km, Narmada 1,312 km.
- Anchor any descending-order question on that four-river ladder.

The Indus system and the Indus Waters Treaty

The Indus rises near the Bokhar Chu glacier in the Kailash range of Tibet and enters India in Ladakh. It is the only major Indian river that flows mostly outside India today, which is exactly why its waters had to be shared by treaty. The treaty split the system geographically - three eastern rivers to India, three western rivers to Pakistan.

- Himalayan tributaries of the Indus - Zaskar, Shyok, Nubra, Hunza, Gilgit.
- Jhelum rises from a spring at Verinag in Anantnag and flows through Wular lake.
- Chenab is formed by the union of the Chandra and the Bhaga at Tandi in Himachal Pradesh.
- Indus Waters Treaty signed at Karachi on 19 September 1960 by Jawaharlal Nehru and Ayub Khan.
- The World Bank is a signatory to the treaty; the Permanent Indus Commission implements it.
- Rajasthan link - the Indira Gandhi Canal draws on the Satluj-Beas share that this treaty gave India.

Who gets which rivers

Group	Rivers	Allotted to
Eastern rivers	Ravi, Beas, Satluj	India
Western rivers	Indus, Jhelum, Chenab	Pakistan

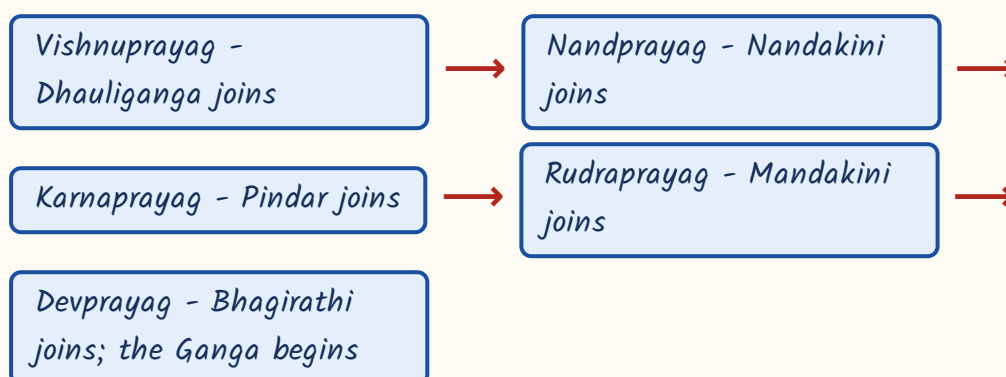
CURRENT LINK

Current link (as of September 2026): India placed the Indus Waters Treaty in abeyance in April 2025 following the Pahalgam terror attack, and has since reaffirmed that position; the dispute is before the Permanent Court of Arbitration. For prelims, the safe core remains the treaty facts - 1960, Karachi, World Bank, eastern rivers to India - with the abeyance as the current-affairs tail.

The Ganga - headwaters, prayags, tributaries

The Ganga is not born as the Ganga. The Bhagirathi comes from Gaumukh at the snout of the Gangotri glacier, the Alaknanda comes from Satopanth, and the two meet at Devprayag - only below that point is the river called the Ganga. It enters the plains at Haridwar. Along the Alaknanda are five confluences, the Panch Prayag, and RPSC has asked them as an upstream-to-downstream ordering question.

Panch Prayag on the Alaknanda, upstream to downstream



Ganga tributaries by bank

Bank	Tributaries
Left bank	Ramganga, Gomti, Ghaghara, Gandak, Burhi Gandak, Kosi, Mahananda
Right bank	Yamuna, Son, Punpun

- Yamuna rises from the Yamunotri glacier on Bandarpunch; its longest tributary is the Tons.
- Peninsular tributaries of the Yamuna - Chambal, Sind, Betwa, Ken. The Chambal is Rajasthan's river.
- Ghaghara is the Karnali in Nepal; the Rapti and the Sarda (Kali) are its tributaries.
- Kosi, the 'Sorrow of Bihar', is formed by the Sun Kosi, Arun and Tamur, jointly the Sapta Kosi.
- Damodar, the 'Sorrow of Bengal', drains the Chotanagpur plateau and joins the Hooghly.
- Damodar Valley Corporation (1948) was India's first multipurpose river valley project.

TRAP

The Son is a RIGHT-bank tributary of the Ganga, not a left-bank one - it rises at Amarkantak and comes up from the south. RPSC has used exactly this in a 'which is NOT a left-bank tributary' question. Only the Yamuna, Son and Punpun join from the right - the three that come up from the south. Everything else on the standard list - Ramganga, Gomti, Ghaghara, Gandak, Kosi, Mahananda - comes down from the Himalaya on the left.

The tributary table RPSC keeps coming back to

This is the highest-value table in the chapter. The same four pairs have been recycled across papers, and the format is always the same - a small tributary in List I, four main rivers in List II. Learn the table, not the reasoning.

Tributary to main river - the documented repeats

Tributary	Joins
<i>Rapti</i>	<i>Ghaghara</i>
<i>Teesta</i>	<i>Brahmaputra</i>
<i>Hiran</i>	<i>Narmada</i>
<i>Purna (Marathwada)</i>	<i>Godavari</i>
<i>Tons</i>	<i>Yamuna</i>
<i>Chambal, Sind, Betwa, Ken</i>	<i>Yamuna</i>
<i>Sabari, Indravati, Manjra, Pranhita</i>	<i>Godavari</i>
<i>Bhima, Koyna, Tungabhadra, Musi</i>	<i>Krishna</i>
<i>Kabini, Hemavati, Bhavani, Amaravati</i>	<i>Kaveri</i>
<i>Tawa (largest), Burhner, Banjar, Orsang</i>	<i>Narmada</i>
<i>Girna, Panjhra, Waghur, Bori</i>	<i>Tapi</i>
<i>Subansiri, Kameng, Manas, Dhansiri, Kopili</i>	<i>Brahmaputra</i>

ASKED IN RAS

Asked in RAS Pre 2021 and 2024 as a four-pair match - Rapti to Ghaghara, Teesta to Brahmaputra, Hiran to Narmada, Purna to Godavari. The individual pairs have also been asked as single one-liners in 2024. If you memorise one table in this chapter, make it this one.

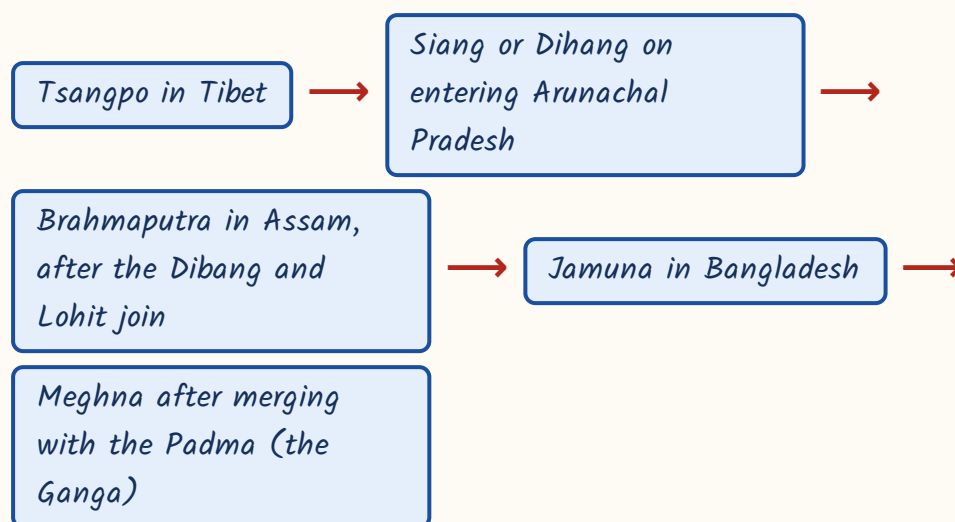
TRAP

There are two Purnas. One is a tributary of the Tapi. The other, which rises in the Ajanta hills of Chhatrapati Sambhajnagar (Aurangabad) district, joins the Godavari - and that is the one the exam has asked. Read the descriptor in the question before you answer. Second pair to guard: the Sabari belongs to the Godavari, not the Krishna - that has been the wrong option in a NOT-correctly-matched question.

The Brahmaputra

The Brahmaputra changes its name four times on its way to the sea, and the name sequence is what gets asked. It also carries an enormous silt load, which is why it braids across the Assam valley and why it built Majuli, the largest river island in India.

The Brahmaputra's names, source to sea



- Tributaries - Subansiri (antecedent), Kameng, Manas, Sankosh, Teesta, Dhansiri, Kopili.
- Majuli in Assam is the largest river island of India.
- The Brahmaputra valley of Assam is among India's most flood-prone regions.

ASKED IN RAS

Asked in RAS Pre 2021 - the Teesta is a tributary of which river? Answer: the Brahmaputra. Do not let the Sikkim location pull you towards the Ganga; the Teesta joins the Brahmaputra in Bangladesh.

TRAP

In Bangladesh the Brahmaputra is the Jamuna, and the Ganga is the Padma. Below their junction the combined stream is the Meghna. RPSC has set 'the Brahmaputra is called the Padma in Bangladesh' as the false statement.

Peninsular rivers - west-flowing and east-flowing

The peninsula is tilted eastwards, so most of its rivers run to the Bay of Bengal. Only a handful run west, and those are the exceptions worth learning first. The Narmada and the Tapi are special because they do not flow in ordinary valleys at all - they occupy rift valleys, downfaulted troughs, and a rift valley river carries its silt straight out to sea instead of spreading it. That is why they make estuaries in the Gulf of Khambhat and not deltas.

- Narmada (1,312 km) rises at Amarkantak and flows west between the Vindhya and the Satpura.
- Dhuandhar falls and the Marble Rocks gorge at Bhedaghat are near Jabalpur on the Narmada.
- Tapi (724 km) rises at Multai in Betul district of Madhya Pradesh, also in a rift valley.
- Mahi is the only Indian river that crosses the Tropic of Cancer twice.
- Sabarmati and Luni both rise in the Aravalli.
- Luni turns saline below Balotra and ends in the Rann of Kachchh - Rajasthan's own west-flowing river.
- Godavari (1,465 km), the 'Dakshin Ganga', rises at Trimbakeshwar near Nashik.
- Krishna (1,400 km) rises near Mahabaleshwar; Kaveri (800 km) at Talakaveri in the Brahmagiri hills of Kodagu, Karnataka.

- Mahanadi rises near Sihawa in Dhamtari district of Chhattisgarh and is dammed at Hirakud.
- Odisha's other rivers - Subarnarekha, Brahmani, Baitarani.

River to place of origin - a standard match

River	Rises at
Godavari	Trimbakeshwar, Nashik
Krishna	Near Mahabaleshwar
Kaveri	Talakaveri, Brahmagiri hills, Kodagu
Mahanadi	Sihawa, Dhamtari, Chhattisgarh
Tapi	Multai, Betul, Madhya Pradesh
Narmada and Son	Amarkantak

Waterfalls - river and state

Waterfall	River	State
Kunchikal	Varahi	Karnataka
Jog (Gersoppa)	Sharavathi	Karnataka
Dudhsagar	Mandovi	Goa
Chitrakote	Indravati	Chhattisgarh
Hundru	Subarnarekha	Jharkhand
Athirappilly	Chalakudy	Kerala
Shivanasamudram	Kaveri	Karnataka
Nohkalikai	-	Meghalaya

TRAP

Two standard traps here. One - the Narmada and the Tapi do NOT build deltas; they form estuaries because they flow through rift valleys. Two - Chitrakote falls are in Chhattisgarh on the Indravati, not in Odisha. Chitrakote is the widest fall in India.

Lakes, Ramsar sites and river interlinking

Lake questions are almost entirely superlative-plus-state. Learn which lake holds which record, and where it is. Note how many of these superlatives Rajasthan owns - Sambhar is India's largest inland saline lake and Keoladeo was one of the country's first two Ramsar sites, so RPSC has a permanent reason to keep coming back to this section.

Lakes - the record and the state

Lake	Record	State / UT
Wular	Largest freshwater lake of India; lies on the Jhelum	Bandipora, J&K
Vembanad	Longest lake in India	Kerala
Chilika	Largest brackish-water lagoon; holds Nalabana Bird Sanctuary	Odisha
Pulicat	Second largest lagoon; Sriharikota lies behind it	Andhra Pradesh - Tamil Nadu
Loktak	Floating phumdi mats; Keibul Lamjao, the only floating national park	Manipur
Kolleru	Lies between the Krishna and Godavari deltas	Andhra Pradesh
Sambhar	Largest inland saline lake of India; a Ramsar site	Rajasthan
Lonar	Meteorite-impact crater lake in Deccan basalt	Buldhana, Maharashtra
Roopkund	The 'Skeleton Lake', about 5,029 m, below Trisul	Chamoli, Uttarakhand
Kanwar (Kabartal)	Large freshwater oxbow lake	Begusarai, Bihar

YAAD RAKHO

Four superlatives, four different words - largest FRESHWATER is Wular, LONGEST is Vembanad, largest BRACKISH lagoon is Chilika, largest inland SALINE is Sambhar. The examiner changes only the adjective. Read the adjective.

- Ramsar Convention signed in 1971 at Ramsar, Iran; India acceded in 1982.
- Chilika (Odisha) and Keoladeo Ghana (Rajasthan) were designated in 1981 as India's first Ramsar sites.
- Sundarban Wetland is India's largest Ramsar site; Renuka in Himachal Pradesh is among the smallest.
- Tamil Nadu has the largest number of Ramsar sites among Indian states.
- Keoladeo and Sambhar are Rajasthan's Ramsar sites - both have been asked directly by RPSC.

ASKED IN RAS

Asked in RAS Pre 2016 and again in 2023 - which two wetlands were designated India's first Ramsar sites in 1981? Answer: Chilika and Keoladeo Ghana. Note that India acceded to the Convention only in 1982, a year after these two were designated - that gap is itself examinable.

CURRENT LINK

Current link (as of September 2026): India's Ramsar tally has risen to 101 sites, the most recent being Glaw Lake in Arunachal Pradesh, designated on 3 August 2026. The count moves, so carry the as-of date; the fixed facts - 1971 Ramsar Iran, India acceded 1982, first sites Chilika and Keoladeo in 1981 - are what a prelims question can safely rest on.

- National Perspective Plan (1980) proposes 30 links - 16 peninsular and 14 Himalayan.

- National Water Development Agency (NWDA) was set up in 1982.
- Ken-Betwa is the first interlinking project taken up; it involves the Daudhan dam, close to the Panna Tiger Reserve.
- Inter-State River Water Disputes Act, 1956 governs the tribunals.
- Tribunals so far - Kaveri, Krishna, Godavari, Narmada (award 1979), Ravi-Beas (Eradi), Mahadayi, Vansadhara.

REVISION RECAP

1. Himalayan rivers are perennial, antecedent and gorge-cutting; peninsular rivers are seasonal with broad graded valleys.
2. Ganga basin is India's largest; Godavari basin the largest peninsular one and the Godavari the longest peninsular river.
3. Indus rises near Bokhar Chu in the Kailash range; Jhelum at Verinag; Chenab from Chandra plus Bhaga at Tandi.
4. Indus Waters Treaty 1960 at Karachi, Nehru and Ayub Khan, World Bank a signatory, Permanent Indus Commission implements it.
5. Eastern rivers Ravi-Beas-Satluj to India; western Indus-Jhelum-Chenab to Pakistan; India held the treaty in abeyance from April 2025.
6. Ganga formed at Devprayag from the Bhagirathi and Alaknanda; enters the plains at Haridwar.
7. Panch Prayag order: Vishnuprayag, Nandprayag, Karnaprayag, Rudraprayag, Devprayag.
8. Right-bank tributaries of the Ganga are only Yamuna, Son and Punpun; the rest are left-bank.
9. Repeat table: Rapti to Ghaghara, Teesta to Brahmaputra, Hiran to Narmada, Purna to Godavari.
10. Brahmaputra names: Tsangpo, Siang/Dihang, Brahmaputra, Jamuna, Meghna; Majuli is India's largest river island.
11. Narmada from Amarkantak and Tapi from Multai flow in rift valleys and make estuaries, not deltas.
12. Mahi crosses the Tropic of Cancer twice; Luni rises in the Aravalli and turns saline below Balotra.
13. Kosi is the Sorrow of Bihar, Damodar the Sorrow of Bengal; DVC 1948 was India's first multipurpose project.
14. Jog on the Sharavathi, Kunchikal on the Varahi, Dudhsagar on the Mandovi, Chitrakote on the Indravati, Hundru on the Subarnarekha.

15. Wular largest freshwater, Vembanad longest, Chilika largest brackish lagoon, Sambhar largest inland saline.
16. Loktak carries phumdis and Keibul Lamjao; Lonar is a meteorite crater lake; Roopkund the Skeleton Lake in Chamoli.
17. Ramsar 1971 Iran, India acceded 1982, first sites Chilika and Keoladeo in 1981, Sundarban largest, Tamil Nadu has the most.
18. NPP 1980 with 30 links, NWDA 1982, Ken-Betwa the first project with the Daudhan dam.